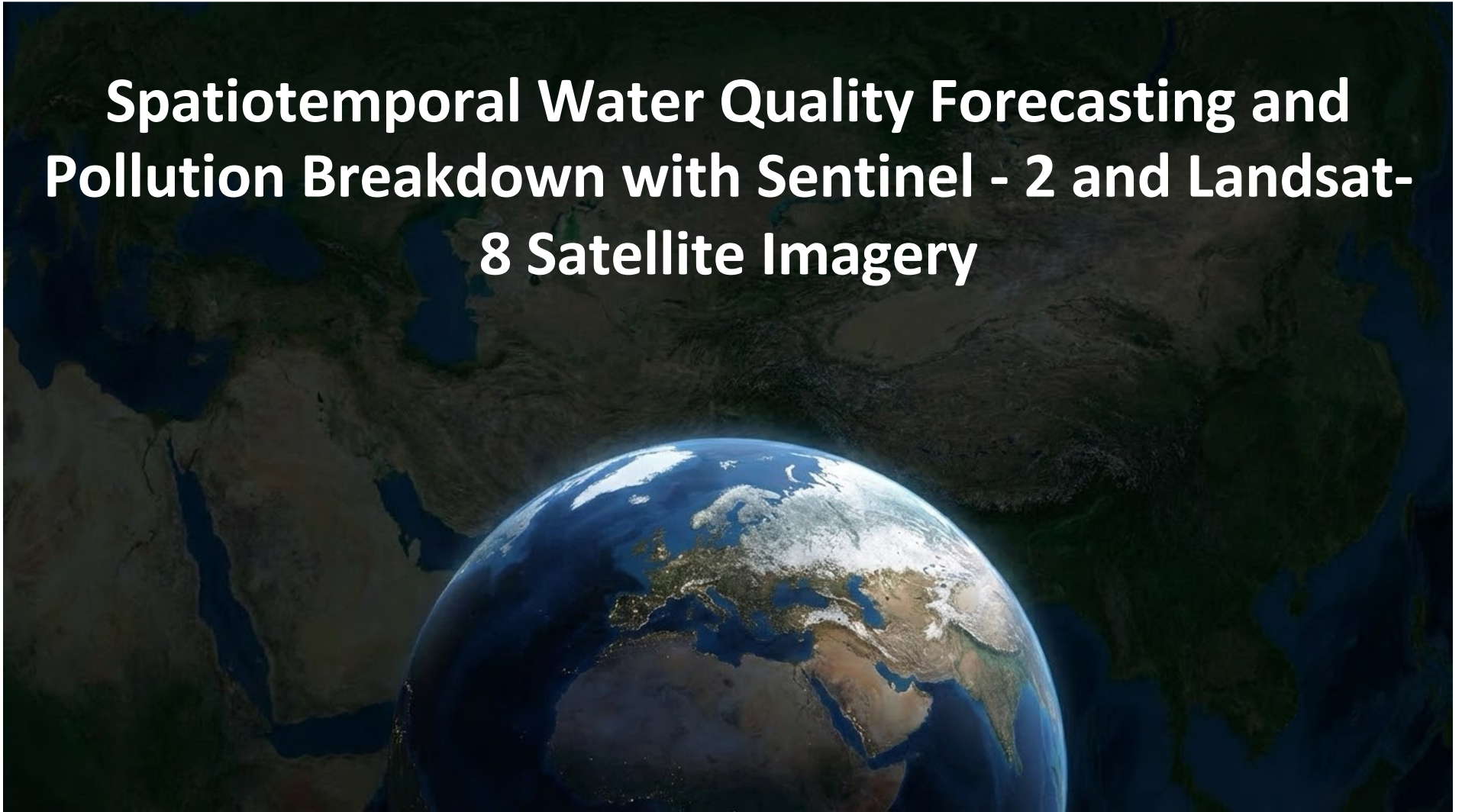


Spatiotemporal Water Quality Forecasting and Pollution Breakdown with Sentinel - 2 and Landsat- 8 Satellite Imagery





Adyar River facing severe water quality degradation

Urbanization and industrial discharge degrade river health
Point-based monitoring fails to capture spatial variation across 128 km



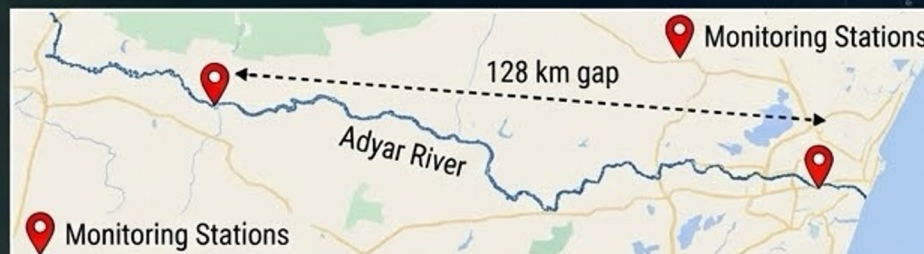
Dense urban areas and industrial zones along Adyar

Large treatment plants, chemical industries, textile mills contribute to degradation
Urban stormwater runoff introduces heavy metals and nutrients
Agricultural runoff from upstream regions adds pesticides and fertilizers



Seasonal fluctuations in water quality parameters

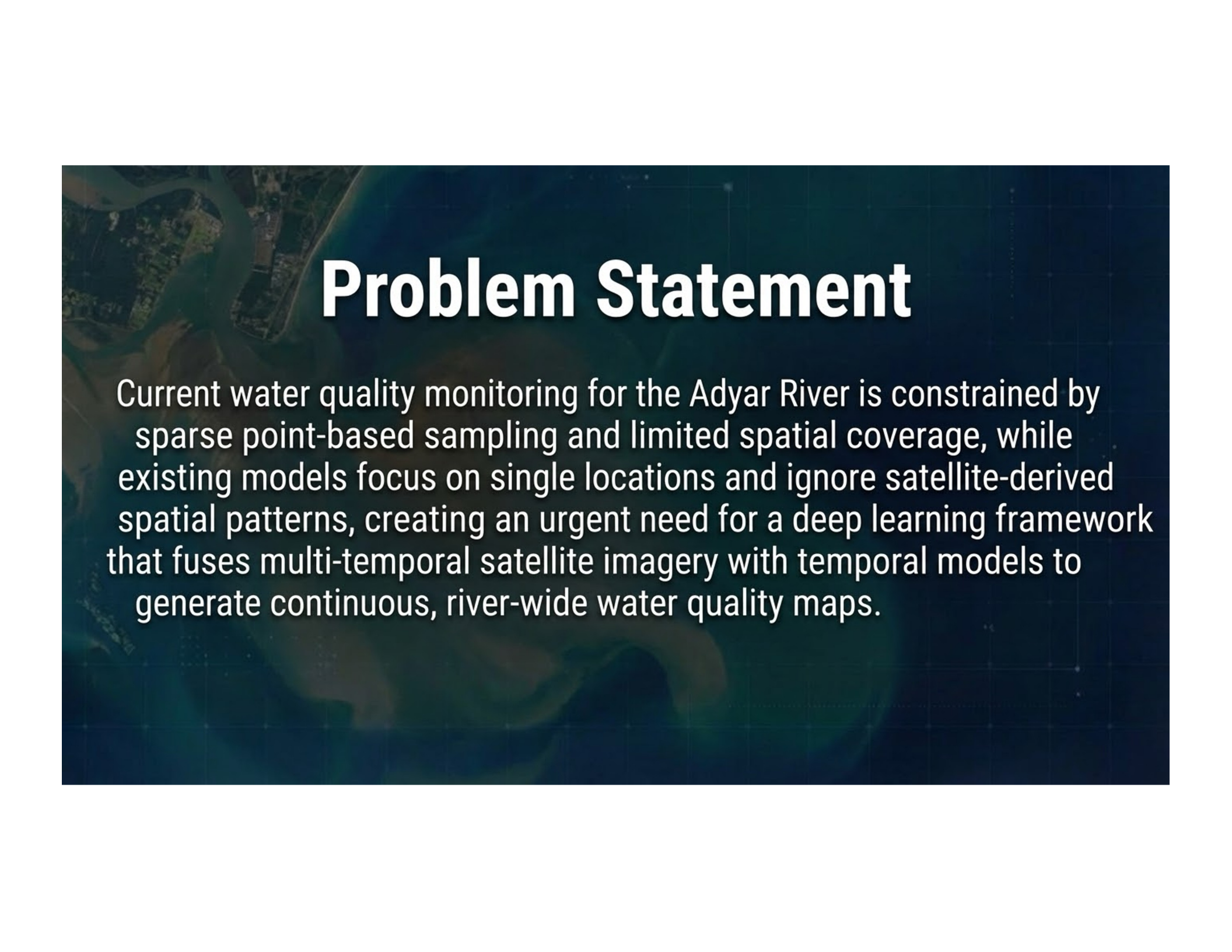
- **Monsoon:** Increased turbidity, high discharge, flooding
- **Summer:** Low water levels, concentrated pollutants, algal blooms
- **Winter:** Stabilized flow, moderate parameters



Current point-based monitoring is insufficient

- Only 1–3 fixed monitoring stations across 128 km river
- Spatial heterogeneity between stations remains unmeasured
- Sampling is infrequent (weekly/monthly), missing rapid pollution events
- Need for continuous, spatially distributed monitoring

Challenge: Traditional point-based water quality monitoring cannot capture the spatial complexity and temporal dynamics of river degradation across the Adyar River system. We need spatially continuous, automated, and scalable solutions to enable effective river management and conservation.



Problem Statement

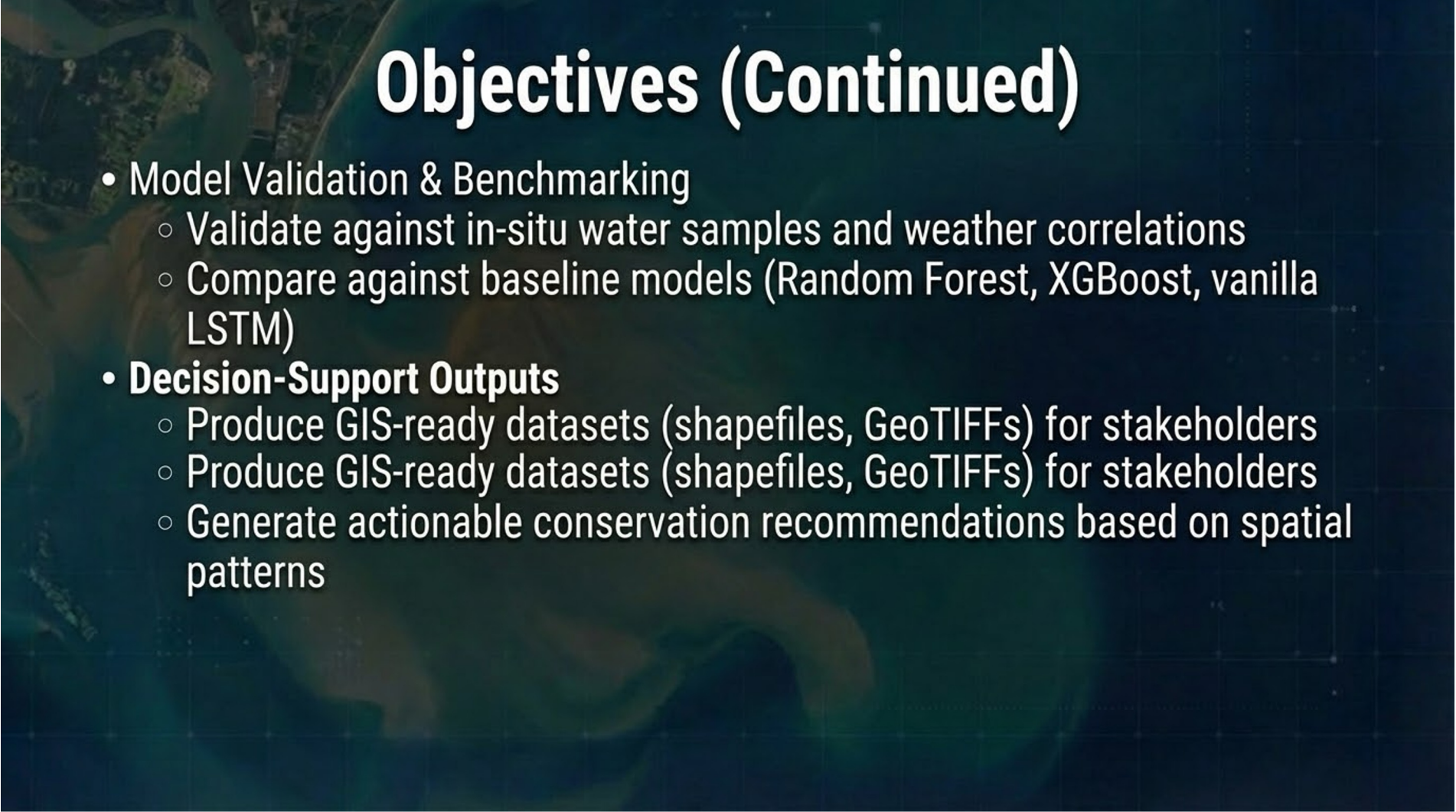
Current water quality monitoring for the Adyar River is constrained by sparse point-based sampling and limited spatial coverage, while existing models focus on single locations and ignore satellite-derived spatial patterns, creating an urgent need for a deep learning framework that fuses multi-temporal satellite imagery with temporal models to generate continuous, river-wide water quality maps.

Objectives

Primary Objective: Develop a spatiotemporal deep learning framework that integrates satellite imagery (Sentinel-2, Landsat-8) with CEEMDAN-CNN-LSTM to generate continuous, river-wide water quality maps for the Adyar River.

Specific Objectives:

- Spatial Extension of Temporal Models
 - Extend Phase 1 point-based predictions to multi-patch spatial inference across the river
- Satellite Data Integration
 - Collect & preprocess 5-year historical archive (2021–2025) of satellite imagery
 - Implement automated cloud masking and spectral index calculation
- Spatial Mapping & Anomaly Detection
 - Generate monthly/seasonal water quality maps across the entire river
 - Identify pollution hotspots and inter-annual trends



Objectives (Continued)

- **Model Validation & Benchmarking**
 - Validate against in-situ water samples and weather correlations
 - Compare against baseline models (Random Forest, XGBoost, vanilla LSTM)
- **Decision-Support Outputs**
 - Produce GIS-ready datasets (shapefiles, GeoTIFFs) for stakeholders
 - Produce GIS-ready datasets (shapefiles, GeoTIFFs) for stakeholders
 - Generate actionable conservation recommendations based on spatial patterns

An aerial photograph of a river, likely the Adyar River, with a grid overlay. The river is dark blue, and the surrounding land is green and brown. The grid is composed of small squares, some of which are highlighted in a lighter shade of blue.

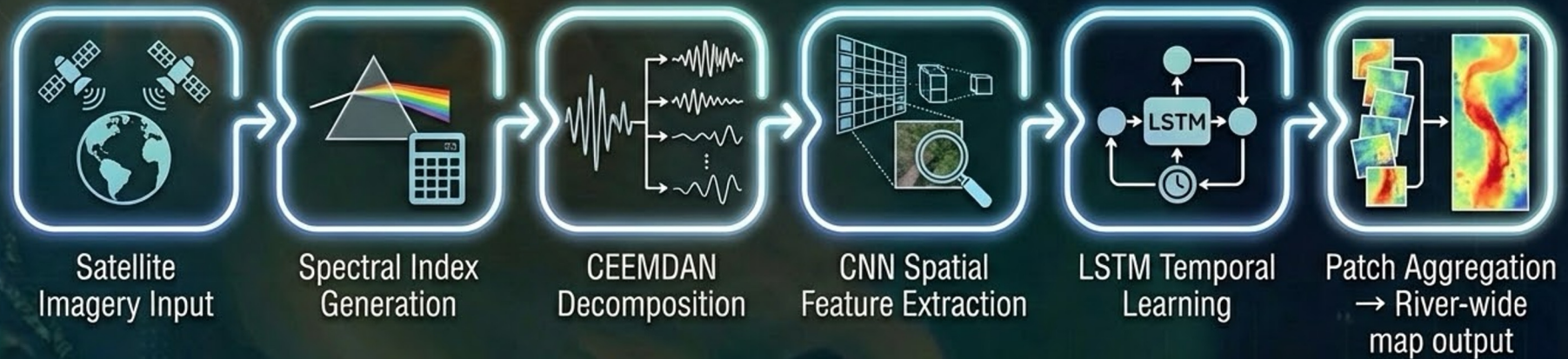
Proposed Solution

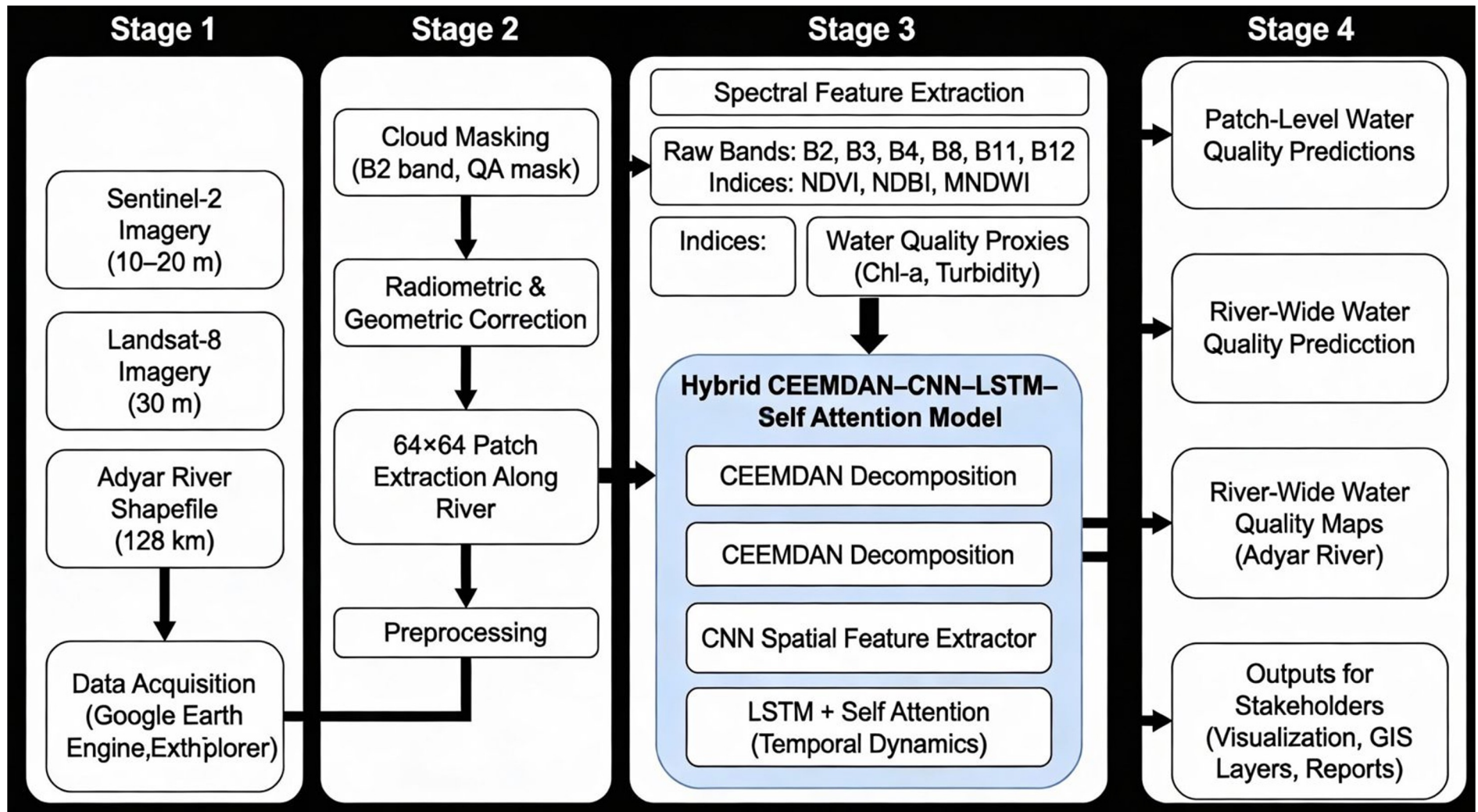
Core Approach: Integrate satellite imagery (Sentinel-2, Landsat-8) with a hybrid CEEMDAN-CNN-LSTM deep learning model to generate spatially continuous water quality predictions across the Adyar River.

Key Components:

- Multi-Spectral Satellite Data
 - Leverage free, cloud-masked Sentinel-2 (10–20m) and Landsat-8 (30m) imagery
 - Extract spectral bands: Blue, Green, Red, NIR, SWIR
- Spectral Index Calculation
 - Derive water quality proxies: NDVI, NDBI, MNDWI, Chlorophyll Index, Turbidity Index
 - Apply domain-specific formulas to convert band ratios to environmental parameters
- Deep Learning Inference
 - Apply Phase 1 CEEMDAN-CNN-LSTM-Attention model per 64×64 spatial patch
 - Generate pixel-level water quality predictions across entire river corridor
- River-Wide Water Quality Maps
 - Aggregate patch predictions into continuous, seasonal water quality maps
 - Identify spatial anomalies and pollution hotspots.

Architecture Diagram





Phase 2 Architecture (Detailed)

Data Acquisition

1. Satellite Image Acquisition

- Download Sentinel-2 & Landsat-8 imagery via Google Earth Engine / USGS EarthExplorer
- 5-year historical archive (2021–2025) covering entire Adyar River corridor

Preprocessing

2. Cloud Masking & Preprocessing

- Detect and remove clouds using B2 band (bright signature) and quality assurance masks
- Perform radiometric and geometric corrections for consistency

3. Spatial Patch Extraction

- Overlay 64×64 pixel grid along river corridor (128 km coverage)
- Extract individual patches for independent processing

Feature Extraction

4. Spectral Index Calculation

- Compute NDVI (vegetation), NDBI (built-up), MNDWI (water extent)
- Derive water quality proxies: Chlorophyll Index (B5/B4), Turbidity Index (Red/Blue ratios)

Inference

5. Deep Learning Inference

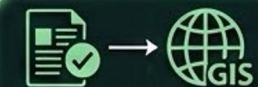
CEEMDAN-CNN-LSTM

- Apply CEEMDAN-CNN-LSTM model to each patch independently
- Generate patch-level water quality predictions

Outputs

6. Spatial Aggregation & Mapping

- Interpolate patch predictions to continuous river-wide maps
- Generate monthly, seasonal, and annual water quality layers



7. Output Generation

- Produce GIS-ready shapefiles and GeoTIFFs
- Create visualizations for stakeholder reports and conservation planning

Tools & Software Used

Satellite & Remote Sensing Platforms



Google Earth Engine (GEE)

Image processing, cloud masking, spectral index calculation



USGS EarthExplorer

Landsat-8 Level-2 data acquisition



Copernicus Hub

Sentinel-2 Level-2A imagery download



SAGA GIS

Raster processing and geospatial analysis

Programming & Deep Learning



Python (Jupyter/Google Colab)

Core development environment



TensorFlow/PyTorch

Deep learning framework for CEEMDAN-CNN-LSTM implementation



GeoPandas

GeoPandas/Rasterio

Geospatial data processing and I/O

Visualization & Analysis



QGIS

Map visualization, spatial analysis, GIS layer generation



Matplotlib/Seaborn

Result plotting and statistical visualization



Google Sheets

Data documentation and collaborative tracking

Dataset



Satellite Imagery

Sentinel-2 (Copernicus)

- **10–20m** resolution
- Blue (B2), Green (B3), Red (B4), NIR (B8), SWIR (B11, B12) bands

Landsat-8 (USGS/NASA)

- **30m** resolution
- Blue, Green, Red, NIR, SWIR bands + Thermal bands

- **Historical Archive:** **5-year** time series (2021–2025) for seasonal and inter-annual analysis

Derived Datasets



- **Spectral Indices:** NDVI, NDBI, MNDWI (computed from raw bands)

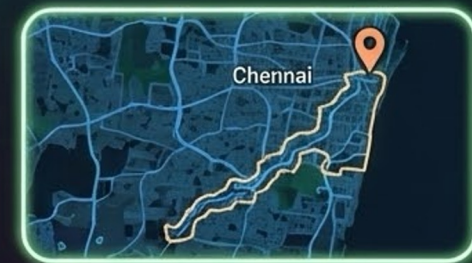


- **Water Quality Proxies:** Chlorophyll-a Index (B5/B4), Turbidity Index (Red/Blue ratios), Suspended Matter indices



- **Ground Truth (Optional):** In-situ water samples from Sandhya ma'am's geotagged dataset with weather correlations

River Delineation



- **Adyar River Shapefile:** **128 km** corridor boundary (from OSM / manual digitization)



- **Spatial Resolution:** **64×64 pixel** patches (~1–2 km coverage per patch)



- **Total Coverage:** **~100–120 patches** across entire river length

Literature Survey

Paper 1 – Drought Monitoring via Spatiotemporal Fusion

- **Title:** A Deep Learning Model for Integrating Landsat-8 and Sentinel-2 Satellite Images...
- **Authors:** Aravinth J and Anand R (2026)
- **Summary:** Enhances drought forecasting resolution by fusing Landsat-8 (spatial) and Sentinel-2 (temporal) data.
- **Methodology:** * Multiscale attention-based fusion mechanism.
 - Bi-directional Long Short-Term Memory (Bi-LSTM) network to predict SPEI-12.
- **Outcomes:** Fused dataset achieved higher correlations ($r=0.72$) than single sensors. The Bi-LSTM model reached an R^2 of 0.7946.

Paper 2 – 40-Year Aquaculture Evolution in Hongze Lake

- **Title:** Analysis of Spatiotemporal Changes in Enclosed Aquaculture Areas in Hongze Lake, China...
- **Authors:** Peng Wang, et al. (2025)
- **Summary:** Investigates the impact of 40 years of aquaculture expansion on lake water quality (TN, TP, Chla).
- **Methodology:** * Object-oriented extraction via Random Forest using Landsat imagery (1984–2024).
 - Regression analysis between aquaculture area and water environmental factors.
- **Outcomes:** Aquaculture area peaked in 2002 ($\sim 600 \text{ km}^2$). Expansion is significantly correlated with increased TN and Chla concentrations.

Paper 3 – Coastline Extraction in Tidal Regions

- **Title:** A Cross-Calibration Approach for Coastline Extraction With Time Series Satellite Images
- **Authors:** Hua Yang, Jintao Xia, and Jiang Zhang (2025)
- **Summary:** Addresses weak spectral separability between land and water in tidal flats using cross-calibration.
- **Methodology:** * Uses a "relative 0-meter isobath" from high-tide images to calibrate low-tide extractions.
 - Steep Slope Identification Algorithm (SSI-A) for high-accuracy calibration.
- **Outcomes:** Reduced extraction errors by 13m (Sentinel-2) and 18m (Landsat-8).

Paper 4 – Super-Resolution for Groundwater Assessment

- **Title:** Deep Learning-Based Super-Resolution for Enhanced Groundwater Quality Assessment...
- **Authors:** Peng Shu, et al. (2025)
- **Summary:** Upscales satellite imagery to improve the monitoring of urban features affecting groundwater.
- **Methodology:** * CNN architecture upscales imagery to 0.5m high-resolution outputs.
 - Fuzzy Analytic Hierarchy Process (FAHP) to calculate Groundwater Quality Index (GWQI).
- **Outcomes:** Achieved PSNR of 32.4 dB. Found a moderate negative correlation ($r = -0.62$) between GWQI and water level depth.

Paper 5 – Multitask Learning for Satellite Masking

- **Title:** Improving Satellite Imagery Masking Using Multitask and Transfer Learning
- **Authors:** Rangel Daroya, et al. (2025)
- **Summary:** A unified system to mask water, clouds, shadows, and snow in HLS imagery for sediment analysis.
- **Methodology:** * Multitask learning (predicting 5 masks simultaneously).
 - Transfer learning using pre-trained models (Satlas, Prithvi).
- **Outcomes:** 9% F1-score improvement in water identification and 30x faster processing for downstream tasks.

Paper 6 – MHNet: Water Body Segmentation

- **Title:** MHNet: A Masked Hybrid Network for Robust Water Body Segmentation...
- **Authors:** Shuo Wang, et al. (2025)
- **Summary:** Combines CNNs and Transformers to solve segmentation issues in low-contrast and complex terrains.
- **Methodology:** * Hybrid multiscale encoder-decoder with Restormer blocks.
 - Self-supervised Masked Autoencoder (MAE) mechanism for feature integration.
- **Outcomes:** Achieved state-of-the-art mIoU of 94.45%; deployed via Google Vertex AI for large-scale monitoring.

Paper 7 – Urban Water Body & Color Analysis

- **Title:** Refined Monitoring of Water Bodies in Built-Up Areas and Analysis of Water Color...
- **Authors:** Bin Li, et al. (2025)
- **Summary:** Tackles the difficulty of water extraction in cities with heavy building shadows and fragmented water.
- **Methodology:** * Benchmark of four models (UNet, DeepLab, PSPNet, Segformer) on high-res Gaofen data.
 - Geographic detector to analyze drivers of water color change.
- **Outcomes:** Segformer performed best (98.64% accuracy). Industrial location and soil erosion identified as primary water's color drivers.

Paper 8 – UAV-Satellite Integration for DOM

- **Title:** Remote Sensing Dissolved Organic Matter in Freshwater Aquaculture Ponds...
- **Authors:** Guangxin Chen, et al. (2025)
- **Summary:** Monitors Dissolved Organic Matter (DOM) in small ponds by bridging the gap between field samples and satellite pixels.
- **Methodology:** * Spatial mapping integrating centimeter-level UAV data with Sentinel-2.
 - Simulated Annealing-Random Forest (SA-RF) optimization.
- **Outcomes:** Model accuracy improved by 10% compared to using satellite data alone ($R^2=0.84$ for UAV base).

Paper 9 – High Solar Zenith Angle Monitoring

- **Title:** Satellite Retrieval of Water Quality Indicators Under High Solar Zenith Angles
- **Authors:** Yongquan Wang, et al. (2025)
- **Summary:** Enables water quality monitoring (SPM, Chl-a) during dawn and dusk (high SZA).
- **Methodology:** * Evaluation of XGBWQI model on Top-of-Atmosphere (ρ_t), Rayleigh-corrected (ρ_{rc}), and R_{rs} data.
- **Outcomes:** ρ_t -based model was most accurate ($R^2=0.90$ for SPM). Successfully mapped diurnal red tide variations.

Paper 10 – Multi-Satellite Water Quality Inversion

- **Title:** Water Quality Variations in the Lower Yangtze River Based on GA-RF Model...
- **Authors:** Wentao Hu, et al. (2025)
- **Summary:** Monitors Nitrogen (TN) and Phosphorus (TP) in the Yangtze River using three-satellite data fusion.
- **Methodology:** * Joint use of GF-1, Landsat-8, and Sentinel-2 data.
 - Genetic Algorithm-Random Forest (GA-RF) weighted by entropy method.
- **Outcomes:** The joint model outperformed single satellites. Observed an 8.9% decrease in TP and 11.3% decrease in TN from 2018–2023.

Phase 1 Completion Summary

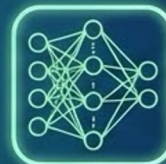
Phase 1 Outputs (Completed):

- Developed **CEEMDAN-CNN-LSTM-Self Attention** hybrid architecture for point-based water quality prediction
- Achieved **temporal pattern recognition** from historical water sample time-series data
- Integrated with **chatbot interface** for user-friendly predictions at single monitoring location
- Validated model performance on **point data** with acceptable accuracy metrics

Key Achievement:

- Proven capability of **hybrid deep learning** to capture complex water quality dynamics
- Foundation for **spatiotemporal extension** to satellite-driven spatial mapping

Phase 1 to Phase 2 Transition

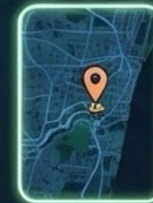


Frozen Model Weights & Transfer Learning

Phase 2 Spatial Extension & Continuity:

- Phase 1 temporal-pattern model provides reusable feature extraction for spatial extension
- Single-point CEEMDAN decomposition + LSTM dynamics are replicated across multiple river patches
- Phase 1 model weights are **frozen** & transferred to Phase 2 for spatial patch predictions
- **Extension: From 1 monitoring point → 128 km river-wide spatial analysis**

Phase 1 (Single Point)



Phase 2 (River-wide)



Phase 2 Workflow

Step 1: Data Preparation

Weeks 1–2

- Finalize Adyar River shapefile with 64×64 grid overlay
- Collect 5-year Sentinel-2 & Landsat-8 data via Google Earth Engine
- Implement automated cloud masking (B2 band thresholding)

✓ Data Ready

Step 2: Spectral Processing

Weeks 3–4

- Calculate NDVI, NDBI, MNDWI for each satellite pass
- Derive chlorophyll & turbidity indices from band ratios
- Generate multi-index stacks per spatial patch

✓ Indices Processed

Step 3: Model Adaptation & Inference

Weeks 5–6

- Transfer Phase 1 CEEMDAN-CNN-LSTM weights to spatial framework
- Apply per-patch inference across entire river corridor
- Aggregate patch predictions to continuous river maps

✓ Model Inference Complete

Step 4: Validation & Analysis

Weeks 7–8

- Validate predictions against in-situ water samples & weather data
- Generate seasonal water quality maps and trend analysis
- Identify spatial anomalies and pollution hotspots

✓ Validation Validated

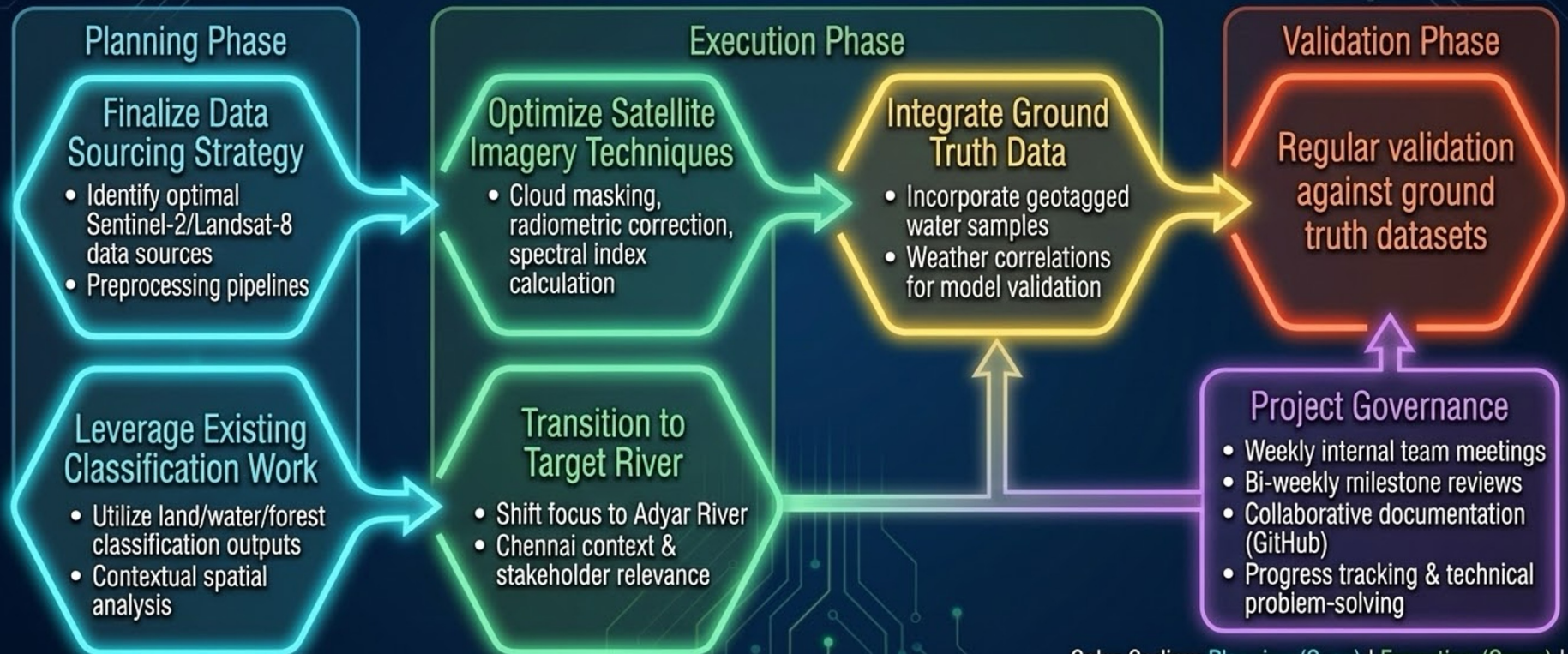
Step 5: Output Generation & Reporting

Weeks 9–10

- Create GIS-ready shapefiles and GeoTIFF layers
- Produce visualization maps and conservation recommendations
- Prepare final report and presentation materials

✓ Final Outputs Delivered

Key Milestones & Project Dependencies



Color Coding: Planning (Cyan) | Execution (Green) | Validation & Critical Path (Orange/Red) | Governance (Purple)

References

- [1] Li, Y., Zhang, Z., & Wang, X. (2024). Satellite-based water quality estimation using Sentinel-2 and Landsat-8 imagery. *Remote Sensing of Environment*, 295, 113698.
- [2] Chen, J., Liu, H., & Wang, X. (2023). Deep learning for spatio-temporal water quality forecasting in urban rivers. *Water Resources Research*, 59(8), e2023WR034891.
- [3] Singh, R., Patel, A., & Kumar, V. (2023). CEEMDAN-LSTM hybrid models for environmental time-series prediction. *IEEE Geoscience and Remote Sensing Letters*, 21, 1–5.
- [4] Copernicus Data Space Ecosystem (2025). Sentinel-2 Level-2A Processing Baseline. Retrieved from <https://dataspace.copernicus.eu/>
- [5] USGS (2025). Landsat 8-9 Collection 2 Level-2 Data. Retrieved from <https://www.usgs.gov/landsat/>
- [6] Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., & Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment*, 202, 18–27.
- [7] Gao, B. C. (1996). NDWI—A normalized difference water index for remote sensing of vegetation liquid water from space. *Remote Sensing of Environment*, 58(3), 257–266.